Different types of batteries based on chemistries (Credit: https://components101.com)

requirements of different components in a robot. Power limits vary depending on the country. In India, the power source is 230 volts, 50Hz alternating current. Energy density is the amount of energy stored in a given system or region per unit volume.

All batteries list a voltage and a capacity rating. Sometimes, a discharge current rating or burst current rating is also mentioned. Balaji Viswanathan, CEO, Invento Robotics, says, "Choice of the battery depends on energy density and safety. We are trying to run batteries for ten to twelve hours for our robots. We inform the users to charge when the charge reaches 25 per cent, and we are also working on an automated docking station to charge at that level."

Voltage rating

The maximum terminal voltage that a battery can deliver to electronic components under ambient conditions is what the voltage rating mentioned on a battery means. Although 9V batteries are usually chosen for electronics, in a robot these cannot be used for the entire system. These can be used only for powering components like microcontrollers and sensors.

If a motor driver IC is being used, then also 9V supply for the logic side is sufficient. Motors draw much more power, so batteries rated AA, AAA, C, D, or battery packs suit the purpose. AAA, AA, C, and D batteries are each rated at 1.5V but vary in size, and the amount of current these can supply increases in ascending order.

Current or capacity rating

The amount of current per hour deliverable by a battery before getting fully discharged is known as its current or capacity rating. It is generally given in amp-hour or milli-Ah (mAh) based on the maximum output current. If a battery has a capacity of xy amp-hours at a current of y ampere, it simply means that y amperes can be provided to a load continuously for x hours.

Capacity helps in determining battery life. Motors need batteries with a high current output for proper functioning. The current rating needs to be more than the sum of the current requirements of individual motors.

C- and E-rates

C- and E-rates for describing batteries can be confusing for a beginner. The discharge current is mostly expressed as a C-rate to normalise against battery capacity. A C-rate measures the rate at which a battery gets discharged relative to its maximum capacity. A 1C rate would imply that the entire battery will take one hour to discharge.

Likewise, an E-rate is a measure of discharge power. A 1E rate is the discharge power needed to completely discharge a battery in one hour.

Battery selection for robots

The factors discussed in the previous section vary according to battery chemistry. There is always a trade-off between power and energy. Only one can be high, not both.

With so many types of robots in the market, selecting batteries can be confusing for a particular application. There needs to be a right balance between the weight of the robot, battery, and choice of motors. Listed next are some commonly used varieties.

Lead-acid batteries. Lead-acid or sealed lead-acid (SLA) batteries are useful for large projects where a large amount of current is needed for longer periods of time. These are cheap and do not require much maintenance. Their voltages are increments of 6V. However, these have low power density, lose capacity when idle, and cannot be used for lightweight mobile robots due to their heavy weight.

Nickel-metal hydride (NiMH). These rechargeable batteries have been a popular choice on the economical side as these can fulfill capacitance and current output for many applications. The voltage rating of one cell is 1.2V, with a range from 600mAh to 3300mAh. Also, these are safer than Ni-Cad batteries since these do not contain cadmium, which is a toxic element. However, these lose charge when idle and are getting replaced by lithium polymer batteries nowadays.

Lithium polymer (Li-Po). These lightweight batteries are often found in electronic items these days. Along with high discharge rates, these are capable of delivering large amounts of power that are many times their capacitance. These batteries are available in a wide voltage range for various needs, in increments of 3.7V. S and P suffixes in ratings are used for series and parallel cells, respectively. However, these batteries need to be handled carefully and require a smart charger.

Lithium ion (Li-ion). These rechargeable batteries (such as 18650 cells) are quite similar to Li-Po but differ in cell construction. These batteries are easily available, have stable voltage while draining and high capacity. These are also lightweight and safe when used with a battery management system. Unlike Li-Po batteries, these are preferred for low current requiring components like controllers and sensors.

Alkaline batteries. These non-rechargeable batteries usually come with 1.5V and 9V ratings. Duracell, Energizer, and Sanyo are some of the